

Title:
Smart Parking & Traffic Management System

Based on the case study below, you are required to propose data structure technique that allows quick retrieval and explain why it is efficient.

Description:

In the Smart Parking & Traffic Management System, the system must manage parking locations and vehicle records efficiently. You are required to design a module that stores and updates parking slot and vehicle information dynamically. The system should support adding, removing, and displaying records. Propose suitable data structures and justify your choice.

The system must also handle vehicles entering and exiting parking areas in real time. Vehicles should be processed in the order they arrive, but the system should also support reversing the most recent action in case of errors. Design a solution using appropriate data structures and explain how it works.

In addition, the system must assign parking slots based on priority, such as nearest available slot or shortest waiting time. You are required to design a mechanism that efficiently selects the best parking slot and explain how your approach improves performance compared to a simple search.

The system should also guide drivers to the nearest available parking location. You are required to represent the parking locations and routes, and implement a method to determine the shortest path. Briefly explain your approach and its efficiency.

Furthermore, the system should allow users to quickly search for vehicle records and parking information. Design a solution that enables efficient searching and sorting, and explain why it performs better than a basic list.

Finally, the system should provide fast access to frequently used data such as vehicle details and parking status.

Marking Rubrics:

Component	Data Structure Technique Proposed	Criteria	Marks
Parking & Vehicle Management	Array / Linked List	Correct DS selection, dynamic data handling (add/delete/display), explanation with complexity	15
Entry & Exit Processing	Queue + Stack	Correct FIFO processing and undo mechanism (LIFO), working logic and justification	15
Parking Slot Assignment	Priority Queue (Min Heap)	Efficient slot selection, correct prioritization, comparison with linear search	15
Route Navigation	Graph + Dijkstra Algorithm	Correct graph modeling, shortest path implementation, explanation of efficiency	20
Search System	BST / AVL Tree	Efficient searching and sorting, correct operations, explanation of $O(\log n)$	15
Fast Data Retrieval	Hash Table (HashMap)	Efficient lookup ($O(1)$), correct key-value usage, explanation	10
System Integration	All DS combined	Modules are integrated and function as one system	5
Presentation & Report	Diagrams & Explanation	Clear report, proper structure, relevant DS diagrams	5
TOTAL			100